

Lab Procedure for Python

Inverse Kinematics

Setup

1. It is recommended that you review [Lab 2 – Application Guide](#) before starting this lab.
2. Turn on the QBot Platform by pressing the power button once. To ensure the robot is ready for the lab, check the following conditions:
 - a. Confirm that LEDs on the robot base are solid red, and the LCD shows a battery voltage above 12.5 V.
 - b. The Logitech F710 joystick's wireless receiver is connected to the QBot Platform. Ensure the switch on the joystick is in the **X position** and that the LED beside the Mode button is **off**.
 - c. Connect your computer to the same network that the QBot Platform is connected to. If using the provided router, connect to the Wi-Fi network **Quanser_UVS-5G**.
3. Test connectivity: Open a command prompt on your local computer, enter the following command, replacing `192.168.x.x` with the QBot's IP address found on its LCD display, and hit enter: `ping 192.168.x.x`

A successful connection returns a line such as: `Reply from 192.168.x.x: bytes=32 time=20ms TTL=64.`
4. In Section A of [inv_kin.py](#), set **ipHost** to your Windows computer's IPv4 address. To find it, run `ipconfig` in Command Prompt and look under your active network connection. Your IP address is listed as the IPv4 address, and should look like the one for the QBot, `192.168.X.X`.

Running Code

You will edit the code on your local Windows machine and then transfer it to the QBot Platform. Repeat the steps in this section each time you change the code. **Refer back to this section later when instructed to run the code.**

1. Open WinSCP, use the QBot's IP address as the **Host name**, and "nvidia" for both Username and Password. Select Login. This opens a connection to the QBot for file transfer.

2. In the WinSCP window, transfer the following files from your local Windows machine to the QBot:
 - a. On the QBot, open
/home/nvidia/Documents/Quanser/6_teaching/3_Robotics/sp1_task_automation/l2_inverse_kinematics/hardware/python, then copy the updated [inv_kin.py](#) and [qbot_platform_driver_physical.rt-linux_qbot_platform](#) into that directory.
 - b. Navigate to the directory
/home/nvidia/Documents/Quanser/0_libraries/python/ shown in Figure 1, or create these directories if they do not exist. Copy over the "hal" and "pal" folder from your local Windows machine, replacing existing files on QBot.

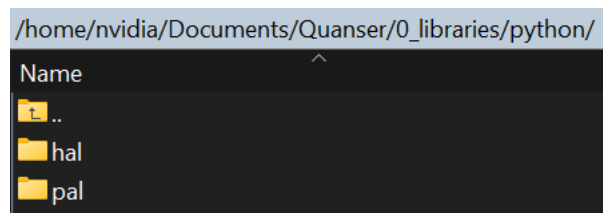


Figure 1: 'hal' and 'pal' folder in correct directory

3. Run [observer.py](#) on your local Windows machine first to initiate receiving data feeds from the QBot Platform. Keep this script running on your Windows machine while you run [inv_kin.py](#) in the next step.
4. Run [inv_kin.py](#) on QBot Platform:
 - a. Open a PuTTY session: at the top menu bar, click Commands > Open in PuTTY.
 - b. In the PuTTY terminal, enter the password "nvidia".
 - c. Navigate to the python script directory from step 2a by using the `cd` command.
 - d. Run the script using the following command:

```
sudo PYTHONPATH=$PYTHONPATH python3 inv_kin.py
```

The LEDs on the base of the QBot will turn blue when the scripts starts successfully.
 - a. To stop both [inv_kin.py](#) and [observer.py](#), press the right button (RB) on the joystick.

Inverse Kinematics Formulation

1. In Section B of [inv_kin.py](#), right click on the **QBPMovement()** object and select "Go to Definition". Verify that **diff_drive_forward_velocity_kinematics()** function has been completed. If not, copy the code from last lab to complete the function.
2. When considering a pure forward velocity for a differential drive robot, what do you expect each wheel to do? How about when turning?
3. When commanding a positive turn velocity for the QBot (counterclockwise), which wheel should spin faster?
4. Navigate to **diff_drive_inverse_velocity_kinematics()** under the **QBPMovement()** class. This function is incomplete and must be complete before use. This function should take body speeds as input and output QBot wheel speeds in rad/s.
5. Complete the **diff_drive_inverse_velocity_kinematics()** function. Save your changes to **QBPMovement()**. Go back to the [inv_kin.py](#) and use the function you completed on the previous step to complete Section D. Note: The joystick commands are mapped to forward speed and turn speed, respectively.
6. Run your code following the steps in the [Running Code](#) section. Drive the QBot. Focusing on the scope display window, how closely do the commanded wheel speeds output from your inverse kinematics function match the measured wheel velocities? Revise your function if needed.
7. How closely do the commanded body velocities match the measured velocities through forward kinematics?
8. When you are satisfied with the output of your function, take a screenshot of the scope windows.
9. What is the key difference between this application and the drive mode in the Play lab?
10. What is the maximum forward or turn speed of the robot you can command? What wheel velocity command does this correspond to?
11. Stop your script using the RB button. Ensure that you save a copy of your completed files for review later.
12. Turn OFF the QBot by single pressing the power button. Do not hold it down. After shutdown, confirm that all the LEDs are off.